HW4

19. Let $\{p_n\}$ be the sequence defined by $p_n = \sum_{k=1}^{n} \frac{1}{k}$. Show that $\{p_n\}$ diverges even though $\lim_{n\to\infty}(p_n - p_{n-1}) = 0$.

2.1.19

$p_n = \sum_{k=1}^{n} \frac{1}{k} = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n}$

$p_{n-1} = \sum_{k=1}^{n-1} \frac{1}{k} = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n-1}$

$p_n - p_{n-1} = \frac{1}{n}$

$\lim_{n\to\infty}(p_n - p_{n-1}) = \lim_{n\to\infty} \frac{1}{n} = 0$

By the p-test, for p-series $\sum_{k=1}^{\infty} \frac{1}{k^p}$, it diverges when $p \le 1$ and converges when $p > 1$

In this problem, $P = 1$

So P_n diverges.

1. Use algebraic manipulation to show that each of the following functions has a fixed point at p precisely when $f(p) = 0$, where $f(x) = x^4 + 2x^2 - x - 3$.

a. $g_1(x) = (3 + x - 2x^2)^{1/4}$

b. $g_2(x) = \left(\frac{x + 3 - x^4}{2}\right)^{1/2}$

c. $g_3(x) = \left(\frac{x+3}{x^2+2}\right)^{1/2}$

d. $g_4(x) = \frac{3x^4 + 2x^2 + 3}{4x^3 + 4x - 1}$

2.2 1.d

if $g_1(p) = p$

$(3 + p - 2p^2)^{\frac{1}{4}} = p$

$3 + p - 2p^2 = p^4$

$p^4 + 2p^2 - p - 3 = 0$

$f(p) = p^4 + 2p^2 - p - 3 = 0$

if $f(p) = 0$

$p^4 + 2p^2 - p - 3 = 0$

$p^4 = -2p^2 + p + 3$

$|p| = (3 + p - 2p^2)^{\frac{1}{4}}$

$\therefore p = (3 + p - 2p^2)^{\frac{1}{4}}$

$\therefore g_1(p) = p \iff f(p) = 0$

10. Use Theorem 2.3 to show that $g(x) = 2^{-x}$ has a unique fixed point on $[\frac{1}{3}, 1]$. Use fixed-point iteration to find an approximation to the fixed point accurate to within 10^{-4}. Use Corollary 2.5 to estimate the number of iterations required to achieve 10^{-4} accuracy and compare this theoretical estimate to the number actually needed.

$g(x) = 2^{-x}$ $[a,b] = [\frac{1}{3}, 1]$

i) $g'(x) = -\ln 2\, 2^{-x} < 0$ $\therefore g(x)$ is monorate decreating and continuous

ii) $g(a) = 2^{-\frac{1}{3}} \approx 0.793 \in [\frac{1}{3}, 1]$ $g(b) = 2^{-1} = 0.5 \in [\frac{1}{3}, 1]$

iii) $g'(x) = -\ln 2 \cdot 2^{-x}$

$|g'(x)| = \ln 2\, 2^{-x}$

$\because 2^{-x}$ is monorately decreasing at $x = \frac{1}{3}$

$\therefore |g'(x)| \le \ln 2 \cdot 2^{-\frac{1}{3}} \approx 0.55 < 1$

$\therefore$ By theorem 2.3, $g(x)$ has a unique fixed point in $[\frac{1}{3}, 1]$

$x_0 = 1$

$x_n = 2^{-x_{n-1}}$

$x_1 = 2^{-1}$

$x_2 = 2^{2^{-1}}$

$\therefore |x_n - x_{n-1}| < 10^{-4}$

By Corollary 2.5,

$|p_n - p| \le \frac{k^n}{1-k} |p_1 - p_0|$

$\frac{k^n}{1-k} |p_1 - p_0| < 10^{-4}$

$\frac{(\ln 2 \cdot 2^{-\frac{1}{3}})^n}{1 - \ln 2 \cdot 2^{-\frac{1}{3}}} |2^{-1} - 1| < 10^{-4}$

$n \approx 15.59 \Rightarrow n = 16$

13. For each of the following equations, determine an interval $[a, b]$ on which fixed-point iteration will converge. Estimate the number of iterations necessary to obtain approximations accurate to within 10^{-5} and perform the calculations.

a. $x = \dfrac{2 - e^x + x^2}{3}$ $[0,1]$

b. $x = \dfrac{5}{x^2} + 2$ $[2.5,3]$

c. $x = (e^x/3)^{1/2}$

d. $x = 5^{-x}$

e. $x = 6^{-x}$

f. $x = 0.5(\sin x + \cos x)$

a. let $g(x) = \frac{2-e^x+x^2}{3}$

$g(0) = \frac{1}{3} \approx 0.33$

$g(1) = \frac{3-e}{3} \approx 0.0939$

$g'(x) = \frac{-e^x+2x}{3}$

$g''(x) = -\frac{e^x}{3} + \frac{2}{3} = 0$

$x = \ln 2 \in [0,1]$

$\therefore g'(x)$ increase at $[0, \ln 2]$ decreases at $[\ln 2, 1]$

$g'(\ln 2) = \frac{-2+2\ln 2}{3} < 0$

$\max_{[0,1]} [g'(x)] = |g'(0)| = \frac{1}{3} \approx 0.33$

let $p_0 = 0.5$ $p_1 = \frac{2-e^{0.5}+\frac{1}{4}}{3}$

$|p_n - p| \le \frac{k^n}{1-k} |p_1 - p_0|$

$\frac{(\frac{1}{3})^n}{1-\frac{1}{3}} \left| \frac{2-e^{0.5}+\frac{1}{4}}{3} - 0.5 \right| \le 10^{-5}$

$n \approx 9.75$

$\therefore n = 10$

b. $g(x) = \frac{5}{x^2} + 2$

$g'(x) = -\frac{10}{x^3}$

$|g'(x)| \le \frac{10}{2.5^3}$ on $[2.5, 3]$

$g(2.5) = \frac{14}{5}$

$g(3) = \frac{23}{9}$

let $p_0 = \frac{11}{4}$ $p_1 = \frac{5}{(\frac{11}{4})^2} + 2$

$|p_n - p| \le \frac{k^n}{1-k} |p_1 - p_0|$

$\frac{(\frac{16}{25})^n}{1-\frac{16}{25}} \times \left| \frac{5}{(\frac{11}{4})^2} + 2 - \frac{11}{4} \right| \le 10^{-5}$

$n \approx 22.6$

$\Rightarrow n = 23$

21. Find a function g defined on $[0, 1]$ that satisfies none of the hypotheses of Theorem 2.3 but still has a unique fixed point on $[0, 1]$.

$$g(x) = \begin{cases} \frac{1}{2} & x = \frac{1}{2} \\ 1 & x \neq \frac{1}{2} \end{cases}$$

$g(x) = x \quad x = \frac{1}{2}$

the unique fixed point is $\frac{1}{2} \in [0,1]$

but $g(x)$ is not a continuous function

22. **a.** Show that Theorem 2.3 is true if the inequality $|g'(x)| \leq k$ is replaced by $g'(x) \leq k$, for all $x \in (a, b)$. [*Hint:* Only uniqueness is in question.]

b. Show that Theorem 2.4 may not hold if inequality $|g'(x)| \leq k$ is replaced by $g'(x) \leq k$. [*Hint:* Show that $g(x) = 1 - x^2$, for x in $[0, 1]$, provides a counter example.]

a. $|g'(x)| \leq k \rightarrow g'(x) \leq k$

Suppose $p, q \in (a,b)$ both are the fixed points.

$g(p) = p \quad g(q) = q$

By MVT, $\exists \xi \in (p,q) \cdot \frac{g(p)-g(q)}{p-q} = g'(\xi)$

$\because g(p) = p \quad g(q) = q$

$\therefore g'(\xi) = 1$

It contradicts to $g'(x) \leq k < 1$ when $x \in (a,b)$

$\therefore$ There can be at most one fixed point.

b. $g(x) = 1 - x^2, \quad x \in [0,1]$

$g'(x) = -2x \leq 0 \leq k$

But $\max_{[0,1]} |g'(x)| = |-2| = 2$

$x = g(x) = 1 - x^2 \quad x^2 + x - 1 = 0$

$\because x \in [0,1] \quad \therefore p = \frac{\sqrt{5}-1}{2}$

$g'(p) = -2p = 1 - \sqrt{5} \approx -1.236$

$|g'(p)| > 1$

$\therefore$ Iterating $x_{n+1} = g(x_n)$ near p is divergent not convergent.

$\therefore$ Theorem 2.4 is not True if $g'(x) \leq k$

The nonlinear equation

$$f(x) := xe^x - 1 = 0 \tag{1}$$

has the solution

$$x^* = 0.56714329040978\ldots$$

Consistent fixed point formulations of $f(x) = 0$ satisfy $G(x) = x$.
Consider the following consistent fixed point formulations of (1):

$$G_1(x) = e^{-x},$$
$$G_2(x) = \frac{1+x}{1+e^x},$$
$$G_3(x) = x + 1 - xe^x.$$

Compute 10 iterations for each of these formulations using the nonlinear fixed point iteration starting from value $x_0 = 0.5$. Measure the error to the fixed point x^*. What do you observe? Which reformulations converge and which do not converge? Explain why some of them converge faster/slower or why they don't converge.
You may use the Matlab scripts and functions available on the bcourses page.

Matlab Scripts.

```
function problem3_fixedpoint()
    x_star = 0.56714329040978;
    x0 = 0.5;
    N = 10;  % number of iterations
    % Define the three fixed point functions
    G1 = @(x) exp(-x);
    G2 = @(x) (1+x) ./ (1+exp(x));
    G3 = @(x) x + 1 - x.*exp(x);
    % Run iterations for each formulation
    [x1, err1] = fixed_point_iter(G1, x0, N, x_star);
    [x2, err2] = fixed_point_iter(G2, x0, N, x_star);
    [x3, err3] = fixed_point_iter(G3, x0, N, x_star);
    % Display results
    fprintf('Iter   G1(x)          Error         G2(x)       Error       G3(x)        Error\n');
    for k = 1:N+1
        fprintf('%2d   %.10f   %.2e   %.10f   %.2e   %.10f   %.2e\n', ...
            k-1, x1(k), err1(k), x2(k), err2(k), x3(k), err3(k));
    end

    % Plot errors
    figure;
    semilogy(0:N, err1, 'o-b', 'LineWidth', 1.5, 'MarkerFaceColor','b'); hold on;
    semilogy(0:N, err2, 's-r', 'LineWidth', 1.5, 'MarkerFaceColor','r');
    semilogy(0:N, err3, 'd-g', 'LineWidth', 1.5, 'MarkerFaceColor','g');
    xlabel('Iteration n'); ylabel('|x_n - x^*|');
    legend('G1(x)=e^{-x}', 'G2(x)=(1+x)/(1+e^x)', 'G3(x)=x+1-xe^x');
    title('Fixed Point Iteration Errors');
    grid on;
end
function [x_vals, errors] = fixed_point_iter(G, x0, N, x_star)
    % Run N iterations of fixed point iteration
    x_vals = zeros(N+1,1);
    errors = zeros(N+1,1);
    x = x0;
    for k = 1:N+1
        x_vals(k) = x;
        errors(k) = abs(x - x_star);
        x = G(x);
    end
end
```

```
>> problem3_fixedpoint
Iter   G1(x)          Error         G2(x)       Error       G3(x)        Error
 0   0.5000000000   6.71e-02   0.5000000000   6.71e-02   0.5000000000   6.71e-02
 1   0.6065306597   3.94e-02   0.5663110032   8.32e-04   0.6756393646   1.08e-01
 2   0.5452392119   2.19e-02   0.5671431650   1.25e-07   0.3478126785   2.19e-01
 3   0.5797030949   1.26e-02   0.5671432904   1.11e-15   0.8553214092   2.88e-01
 4   0.5600646279   7.08e-03   0.5671432904   3.89e-15   -0.1565059554   7.24e-01
 5   0.5711721490   4.03e-03   0.5671432904   3.89e-15   0.9773264227   4.10e-01
 6   0.5648629470   2.28e-03   0.5671432904   3.89e-15   -0.6197642519   1.19e+00
 7   0.5684380476   1.29e-03   0.5671432904   3.89e-15   0.7137130874   1.47e-01
 8   0.5664094527   7.34e-04   0.5671432904   3.89e-15   0.2566266491   3.11e-01
 9   0.5675596343   4.16e-04   0.5671432904   3.89e-15   0.9249206769   3.58e-01
10   0.5669072129   2.36e-04   0.5671432904   3.89e-15   -0.4074224055   9.75e-01
```

